

Transformations of exponential function

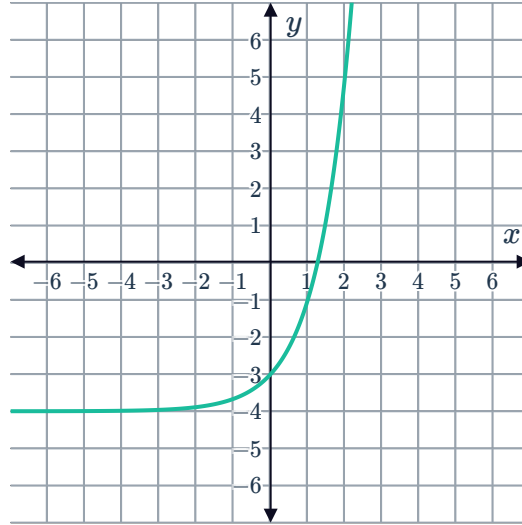


1 $y = 2^x - 7$

2

a Translate the graph downwards by 4 units.

b



3

a 0

b 5

c 0

d $y = 5$

4

a $(0, 1)$

b -1

c $y = 0$

d $y = -2$

5 $g(x) = 2^x - 3$

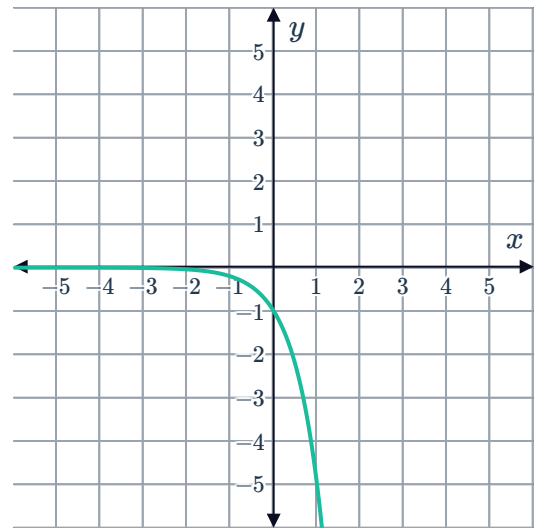
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a $g(x) = f(x + 4)$. A translation of 4 units to the left.

b $g(x) = 3^{x+4}$

7

- a A reflection about the x -axis.



8

- a 4

c $y = 4 \times 2^x$

- b A reflection about the x -axis.

9 $y = 3 \times 2^x$

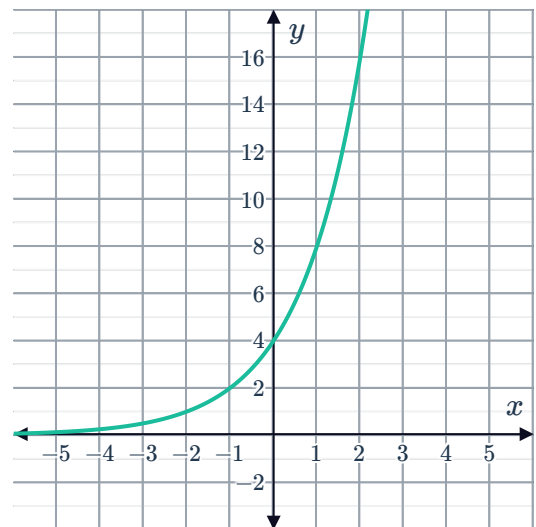
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- a $(0, 4)$

c ∞

- b No

d



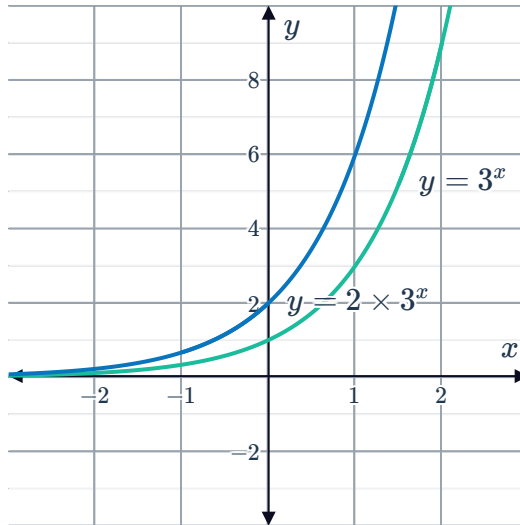
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a

Point on original graph	$\left(-1, \frac{1}{3}\right)$		$(1, 3)$	$(2, 9)$
Point on new graph	$\left(-1, \frac{2}{3}\right)$	$(0, 2)$	$(1, 6)$	$(2, 18)$

b $y = 2 \times 3^x$

c



d Above

e Above

12

a Decreasing

b 0

c $-\infty$

d $(0, -2.5)$